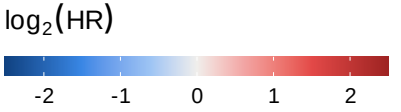
B1 Cause-specific hazard ratios by first-progression pattern (reference: Sys-TG)

All-cause: Cox HR; cause-specific: csHR (cause-specific Cox). Lunn–McNeil heterogeneity across IH/EH/Systemic HCC death: pattern × cause $\chi^2(16) = 60.2$, $P < 0.001$; site × cause $\chi^2(4) = 48.8$, $P < 0.001$; form × cause $\chi^2(4) = 7.7$, $P = 0.104$

	All-cause	Intrahepatic HCC death	Extrahepatic HCC death	Systemic HCC death	Non-HCC death
IH-TG	0.58 (0.41–0.82)**	0.71 (0.44–1.15)	0.88 (0.26–3.04)	0.28 (0.16–0.51)***	NE (sparse reference)
IH-NL	0.50 (0.35–0.70)***	0.64 (0.39–1.05)	0.52 (0.15–1.88)	0.24 (0.13–0.45)***	NE (sparse reference)
IH-TG+NL	0.79 (0.53–1.16)	1.07 (0.63–1.80)	1.32 (0.33–5.30)	0.34 (0.16–0.72)**	NE (sparse reference)
EH-TG	0.91 (0.62–1.34)	1.01 (0.59–1.75)	2.33 (0.65–8.35)	0.49 (0.24–0.98)*	NE (sparse reference)
EH-NL	0.62 (0.44–0.89)**	0.54 (0.33–0.89)*	1.58 (0.47–5.34)	0.47 (0.27–0.84)*	NE (sparse reference)
EH-TG+NL	0.85 (0.50–1.44)	0.73 (0.34–1.59)	2.15 (0.41–11.31)	0.56 (0.22–1.43)	NE (sparse reference)
Sys-TG	Reference	Reference	Reference	Reference	Reference (1 event)
Sys-NL	0.84 (0.56–1.26)	0.66 (0.37–1.20)	3.07 (0.84–11.18)	0.80 (0.42–1.55)	NE (sparse reference)
Sys-TG+NL	0.88 (0.62–1.26)	0.82 (0.50–1.36)	1.69 (0.48–5.90)	0.88 (0.50–1.56)	NE (sparse reference)
Overall Wald P (df=8)	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P < 0.001**	**P = 0.405**



Adjusted for TTP, age, sex, HBV, ECOG PS, ALBI grade, AFP ≥400, PIVKA-II ≥1000, tumour number >3, tumour size, Vp class, HVTT, EHS, cirrhosis; stratified by treatment group. NE: <3 events in the cell or in the reference cell. * P<0.05, ** P<0.01, *** P<0.001 (unadjusted; BH-FDR q values in Table S-HR). Overall Wald test of the 8 pattern terms does not depend on the reference.